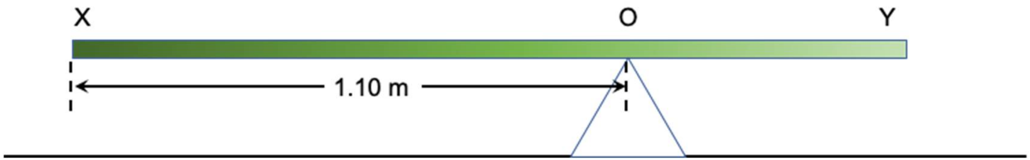
4	A non-uniform rod XY is balanced horizontally on top of a triangular wedge. The top of the wedge O is at a distance 1.10 m from the end X of the rod.



A 1500 g mass is hung at a distance of 0.20 m from X. The rod now needs to be moved 0.25 m to the right to balance horizontally.

What is the mass of the rod?

A	1.2 kg	**B**	3.9 kg	**C**	6.6 kg	**D**	7.8 kg

Answer: B

As the rod is balanced at O, it can be deduced that the centre of mass of the rod is at 1.10 m away from X (at O).

After the extra mass is added, and balanced distance achieved,

Taking moments about point O,
Sum of clockwise moments = Sum of anticlockwise moments
$0.25 \times m \times 9.81 = (1.10 - 0.20 - 0.25) \times (1500 \times 10^{-3}) \times 9.81$
$m = 3.9$ kg